

Skills Summary

Sequences, Series, and the Binomial Theorem

Use this reference while completing your worksheet or when studying.

Skill 1 — Binomial coefficients (binomial-coefficient)

Definition and shortcuts:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Cancel the larger factorial first: $\binom{n}{k}$ is a product of k falling factors over $k!$. Symmetry: $\binom{n}{k} = \binom{n}{n-k}$. Row n of Pascal's triangle lists $\binom{n}{0}$ through $\binom{n}{n}$.

Example: Evaluate $\binom{7}{2}$.

Step 1. Order does not matter here — this is a combination, so use $n!/(k!(n-k)!)$ rather than a permutation count.

Step 2. $\frac{7!}{2!5!} = \frac{7 \cdot 6}{2 \cdot 1} = 21$.

Try it: Evaluate $\binom{9}{3}$.

check: 84

Skill 2 — Expanding a binomial power (full-expansion)

Binomial theorem:

$$(u+v)^n = \sum_{k=0}^n \binom{n}{k} u^{n-k} v^k$$

The exponents of u count down while those of v count up, and the two always sum to n . Whatever stands in the v slot — including a coefficient or a sign — is raised to the power k .

Example: Expand $(x+3)^3$.

Step 1. All four terms are wanted, so run the whole binomial theorem with row 3 of Pascal's triangle: 1, 3, 3, 1.

Step 2. $x^3 + 3x^2(3) + 3x(9) + 27 = x^3 + 9x^2 + 27x + 27$.

Try it: Expand $(x-2)^3$.

check: $8 - 12x + 6x^2 - x^3$

Skill 3 — One specific term (specific-term)

Rule:

To get the coefficient of u^k in $(u + v)^n$, use the *one* term $\binom{n}{k} u^k v^{n-k}$ — do not expand everything.

The coefficient is $\binom{n}{k}$ **times** the power of the constant. Two factors, never just the binomial coefficient.

Example: Find the coefficient of x^2 in $(x + 4)^5$.

Step 1. Only one term carries x^2 , so pick $k = 2$ instead of expanding all six terms.

Step 2. $\binom{5}{2} 4^{5-2} = 10 \cdot 64 = \mathbf{640}$.

Try it: Find the coefficient of x^3 in $(x + 2)^6$.

check: 091

Skill 4 — Structure of a row of coefficients (sequence-structure)

Rules that come from the row itself:

$$\sum_{k=0}^n \binom{n}{k} = 2^n, \quad \sum_{k=0}^n \binom{n}{k} c^k = (1 + c)^n$$

Adding every coefficient of an expansion is the same as substituting 1 for each variable — that is why the sums above are so short.

Example: Add row 4 of Pascal's triangle: $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4}$.

Step 1. This is the sum of a whole row, so use the row-sum rule instead of adding blindly (then check by adding).

Step 2. $2^4 = \mathbf{16}$, and indeed $1 + 4 + 6 + 4 + 1 = 16$.

Try it: Add all the coefficients in the expansion of $(x + 1)^6$.

check: 19

(!) Watch out: $\binom{n}{k}$ alone is the coefficient only when both parts of the binomial are bare, as in $(x + y)^n$. In $(2x - 1)^4$ or $(x + 4)^5$ the constant's power multiplies it, and a minus sign makes every odd- k term negative.